

HYPOTHESIS TESTING

BASIC PRINCIPLES OF HYPOTHESIS TESTING

Definition of Hypothesis Testing

- Hypothesis testing is a statistical method used to make inferences or draw conclusions about a population based on a sample of data.
 - The process involves formulating two competing hypotheses, the null hypothesis (H_0) and the alternative hypothesis (H_1 or H_A), and assessing the evidence from the sample to determine which hypothesis is more supported by the data.
- [Link to Lecture 3](#)

Null and Alternative Hypothesis

- Null Hypothesis (H_0): The null hypothesis represents the default assumption or the status quo.
 - It states that there is no significant difference or relationship between variables or that an observed effect is due to chance.
 - It is usually denoted as H_0 .
- Alternative Hypothesis (H_1 or H_A): The alternative hypothesis is the statement that contradicts the null hypothesis.
 - It represents the researcher's claim or the possibility of a significant difference or relationship between variables.
 - It is denoted as H_1 or H_A .

Examples of Null Hypotheses in Biostatistics

- These examples highlight various scenarios in biostatistics where the null hypothesis represents the absence of an effect, difference, or association between variables or populations.
 - Null Hypothesis: There is no difference in mean blood pressure between patients receiving Drug A and patients receiving a placebo.
 - Null Hypothesis: The proportion of individuals with a specific genetic mutation is the same in males and females.
 - Null Hypothesis: There is no association between smoking status and the incidence of lung cancer.

Examples of Null Hypotheses in Biostatistics

- Null Hypothesis: The mean cholesterol levels in a population before and after a dietary intervention are equal.
- Null Hypothesis: The mean body mass index (BMI) is the same across different age groups.
- Null Hypothesis: The odds of developing a certain disease are equal for individuals exposed and not exposed to a particular environmental factor.

Examples of Null Hypotheses in Biostatistics

- Null Hypothesis: There is no difference in survival rates between two different treatment groups for a specific disease.
- Null Hypothesis: The mean systolic blood pressure is the same among individuals with and without a family history of hypertension.
- Null Hypothesis: The prevalence of a certain disease is not different among individuals of different socioeconomic statuses.
- Null Hypothesis: There is no correlation between dietary intake of a specific nutrient and body weight.

Significance Level

- The significance level, often denoted as α (alpha), is the threshold set by the researcher to determine the level of evidence required to reject the null hypothesis.
 - Commonly used significance levels are 0.05 (5%) and 0.01 (1%).
 - The significance level is a measure of the strength of the evidence that must be present in your sample before rejecting the null.
 - The probability of rejecting the null when it is true.

Test Statistic

- A test statistic is a numerical value calculated from the sample data that helps in assessing the evidence against the null hypothesis.
 - The choice of test statistic depends on the nature of the hypothesis being tested and the type of data being analyzed.

Test Statistic distribution

- Under the assumption that the null hypothesis is true, the test statistic follows a known probability distribution.
- The choice of distribution depends on the type of test being conducted (e.g., t-distribution, normal distribution, chi-square distribution).
- The test statistic distribution is used to calculate the p-value.

Test Statistic distribution

- Under the assumption that the null hypothesis is true, the test statistic follows a known probability distribution.
- The choice of distribution depends on the type of test being conducted
 - **t - distribution** – for calculating CIs for and testing hypothesis about one or two means
 - **Chi-square distribution** – useful for analyzing categorical data
 - **F - distribution** – used for comparing variances and more than two means using ANOVA

p - value

- The p-value is the probability of obtaining a test statistic as extreme as or more extreme than the one observed, assuming that the null hypothesis is true.
 - It measures the strength of evidence against the null hypothesis.
 - If the p-value is smaller than the significance level (α), it is considered statistically significant, and the null hypothesis is rejected in favor of the alternative hypothesis.

Conclusion

- Based on the p-value and the significance level, a decision is made regarding the null hypothesis.
 - If the p-value is less than α , the null hypothesis is rejected, and the alternative hypothesis is supported.
 - If the p-value is greater than or equal to α , there is insufficient evidence to reject the null hypothesis.
- [Link to Lecture 4](#)

Errors in Hypothesis Testing

- There are two types of errors that can occur when making conclusions about the null hypothesis.
 - Type I Error
 - Type II Error

Type I error

- A Type I error occurs when the null hypothesis (H_0) is incorrectly rejected, even though it is true.
 - It is the error of concluding that there is a significant effect or relationship when, in reality, there is no such effect or relationship.
 - Type I error is also known as a "**false positive**."
 - The probability of committing a Type I error is equal to the chosen **significance level** (α) of the test.
 - For example, if α is set at 0.05, it means that there is a 5% chance of committing a Type I error.
- Implication:
 - Committing a Type I error can lead to the rejection of a null hypothesis that is actually true.
 - This can result in incorrect conclusions, wasted resources, and potentially adopting new practices or interventions based on faulty evidence.

Type II error

- A Type II error occurs when the null hypothesis (H_0) is incorrectly accepted, even though it is false.
 - It is the error of failing to reject the null hypothesis when there is a true effect or relationship present.
 - Type II error is also known as a "**false negative**."
 - The probability of committing a Type II error is denoted as β (beta) and is related to the power of the statistical test.
 - Power ($1 - \beta$) represents the probability of correctly rejecting a false null hypothesis.
- Implication:
 - Committing a Type II error means that a real effect or relationship goes undetected or remains unacknowledged.
 - This can lead to missed opportunities, failure to identify important findings, and a lack of action or intervention when it is necessary.

Type I and Type II error

- It's important to note that Type I and Type II errors are inversely related.
- Reducing the probability of one type of error (e.g., decreasing α to reduce Type I error) typically increases the probability of the other type of error (e.g., increasing β and thus increasing the likelihood of Type II error).

Power of a Test

- The chance of making a Type II error is denoted by β (beta);
 - its complement, $(1 - \beta)$, is the power of the test.
- The power, therefore, is the probability of rejecting the null hypothesis when it is false;
 - i.e. it is the chance (usually expressed as a percentage) of detecting, as statistically significant, a real treatment effect of a given size.

Power of a Test

- Ideally, we should like the power of our test to be 100%
 - this is impossible because there is always a chance, albeit slim, that we could make a Type II error.
- Fortunately, however, we know which factors affect power, and thus we can control the power of a test by giving consideration to them.
- It is essential that we know the power of a proposed test at the planning stage of our investigation.
 - Clearly, we should only embark on a study if we believe that it has a 'good' chance of detecting a clinically relevant effect, if one exists (by 'good' we mean that the power should be at least 80%).
 - It is ethically irresponsible, and wasteful of time and resources, to undertake a clinical trial that has, say, only a 40% chance of detecting a real treatment effect.

Factor affecting the Power of a Test

- The sample size:
- Power increases with increasing sample size.
 - A large sample has a greater ability than a small sample to detect a clinically important effect if it exists.
 - When the sample size is very small, the test may have an inadequate power to detect a particular effect.

Factor affecting the Power of a Test

- The variability of the observations: power increases as the variability of the observations decreases
- The effect of interest: the power of the test is greater for larger effects.
 - A hypothesis test thus has a greater chance of detecting a large real effect than a small one.

Factor affecting the Power of a Test

- **The significance level:** the power is greater if the significance level is larger (this is equivalent to the probability of the Type I error (α) increasing as the probability of the Type II error (β) decreases).
 - We are more likely to detect a real effect if we decide at the planning stage that we will regard our P-value as significant if it is less than 0.05 rather than less than 0.01.
- Note that an inspection of the confidence interval for the effect of interest gives an indication of whether the power of the test was adequate.
 - A wide confidence interval results from a small sample and/or data with substantial variability, and is a suggestion of low power.

Module 7: Hypothesis Testing

Module Assignment

Source notes cover null/alternative hypotheses, significance level, test statistics, p-values, Type I/II errors and statistical power.

1. Formulate H_0 and H_1 for three clinical research questions: treatment effect on blood pressure, difference in disease prevalence, and improvement in patient outcome.
2. For each question, state whether a one-tailed or two-tailed approach would be appropriate and justify it.
3. Interpret hypothetical p-values of 0.001, 0.04 and 0.20 at $\alpha = 0.05$.
4. For each result, distinguish statistical significance from clinical importance.
5. Give one clinical consequence of a Type I error and one of a Type II error.

Submission guidance: Show working where calculations are required. Interpret statistical results in clinical language. Use Excel, Jamovi or SPSS where appropriate.